Closed-loop object search | same initial scene, independent policies

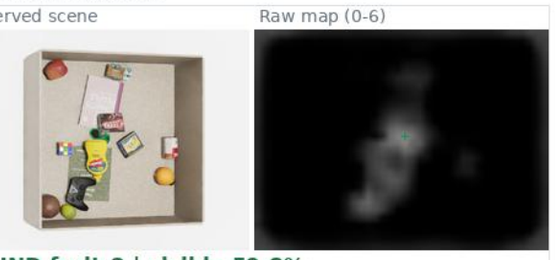
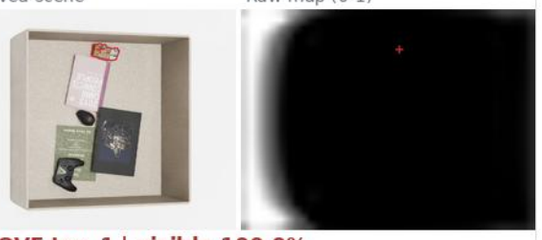
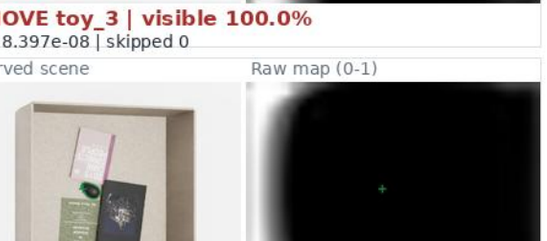
Target: fruit_2 Camera: left Scene: fruit_2_scene00044_env0057

Red: remove blocker. Green: target selected. A row is one policy decision before removal.

Object peak score -> first candidate with visibility >= 50%; step 0 is the common starting image.

Target reference



	Fusion | removals: 1	Similarity | removals: 11	Occlusion | removals: 12	Complexity | removals: 2
r=0	Observed scene / Raw map (0-1)  **REMOVE book_2 | visible 84.1%** score 0.3926 | skipped 0	Observed scene / Raw map (0-1)  **REMOVE fruit_4 | visible 100.0%** score 0.7962 | skipped 0	Observed scene / Raw map (0-1)  **REMOVE packaged_food_3 | visible 100.0%** score 0.3808 | skipped 0	Observed scene / Raw map (0-6)  **REMOVE book_2 | visible 84.1%** score 3.452 | skipped 0
r=1	Observed scene / Raw map (0-1)  **FOUND fruit_2 | visible 60.4%** score 0.9378 | skipped 0	Observed scene / Raw map (0-1)  **REMOVE fruit_1 | visible 100.0%** score 0.7945 | skipped 0	Observed scene / Raw map (0-1)  **REMOVE packaged_food_4 | visible 100.0%** score 0.2911 | skipped 0	Observed scene / Raw map (0-6)  **REMOVE book_1 | visible 81.0%** score 3.829 | skipped 0
r=2	Finished at r=1 No further observation or action	Observed scene / Raw map (0-1)  **REMOVE fruit_3 | visible 100.0%** score 0.7865 | skipped 0	Observed scene / Raw map (0-1)  **REMOVE toy_2 | visible 81.9%** score 0.2343 | skipped 0	Observed scene / Raw map (0-6)  **FOUND fruit_2 | visible 59.8%** score 3.187 | skipped 0
r=3	Finished at r=1 No further observation or action	Observed scene / Raw map (0-1)  **REMOVE packaged_food_2 | visible 100.0%** score 0.5033 | skipped 0	Observed scene / Raw map (0-1)  **REMOVE packaged_food_2 | visible 100.0%** score 0.1899 | skipped 0	Finished at r=2 No further observation or action
r=4	Finished at r=1 No further observation or action	Observed scene / Raw map (0-1)  **REMOVE packaged_food_3 | visible 99.8%** score 0.5021 | skipped 0	Observed scene / Raw map (0-1)  **REMOVE fruit_1 | visible 100.0%** score 0.08682 | skipped 0	Finished at r=2 No further observation or action
r=5	Finished at r=1 No further observation or action	Observed scene / Raw map (0-1)  **REMOVE packaged_food_4 | visible 100.0%** score 0.4996 | skipped 0	Observed scene / Raw map (0-1)  **REMOVE fruit_4 | visible 100.0%** score 0.01887 | skipped 0	Finished at r=2 No further observation or action
r=6	Finished at r=1 No further observation or action	Observed scene / Raw map (0-1)  **REMOVE packaged_food_1 | visible 100.0%** score 0.4984 | skipped 0	Observed scene / Raw map (0-1)  **REMOVE packaged_food_1 | visible 100.0%** score 0.0001457 | skipped 0	Finished at r=2 No further observation or action
r=7	Finished at r=1 No further observation or action	Observed scene / Raw map (0-1)  **REMOVE toy_2 | visible 81.9%** score 0.2069 | skipped 0	Observed scene / Raw map (0-1)  **REMOVE book_2 | visible 100.0%** score 0.0001016 | skipped 0	Finished at r=2 No further observation or action
r=8	Finished at r=1 No further observation or action	Observed scene / Raw map (0-1)  **REMOVE toy_3 | visible 100.0%** score 0.2027 | skipped 0	Observed scene / Raw map (0-1)  **REMOVE fruit_3 | visible 99.9%** score 2.81e-06 | skipped 0	Finished at r=2 No further observation or action
r=9	Finished at r=1 No further observation or action	Observed scene / Raw map (0-1)  **REMOVE book_3 | visible 90.9%** score 0.2021 | skipped 0	Observed scene / Raw map (0-1)  **REMOVE toy_4 | visible 100.0%** score 3.382e-07 | skipped 0	Finished at r=2 No further observation or action
r=10	Finished at r=1 No further observation or action	Observed scene / Raw map (0-1)  **REMOVE book_2 | visible 100.0%** score 0.2008 | skipped 0	Observed scene / Raw map (0-1)  **REMOVE toy_1 | visible 100.0%** score 1.928e-07 | skipped 0	Finished at r=2 No further observation or action
r=11	Finished at r=1 No further observation or action	Observed scene / Raw map (0-1)  **FOUND fruit_2 | visible 99.9%** score 0.9799 | skipped 0	Observed scene / Raw map (0-1)  **REMOVE toy_3 | visible 100.0%** score 8.397e-08 | skipped 0	Finished at r=2 No further observation or action
r=12	Finished at r=1 No further observation or action	Finished at r=11 No further observation or action	Observed scene / Raw map (0-1)  **FOUND fruit_2 | visible 100.0%** score 7.693e-09 | skipped 0	Finished at r=2 No further observation or action

After step 0, removal choices and observed scenes can differ across policies. Blank cells mean the policy has finished.

Grayscale: fixed F/S/O 0-1, C 0-6 display cap; no per-image normalization. Contours/marks use simulator IDs for action annotation.

Simulator visibility eligibility and idealized object removal; target retrieval adds one separate action. Post-hoc extreme examples.